

AREA — Short Revision Notes

Understand Level: Explain Ideas & Concepts

1 What Is Area?

Concept	Explanation	Example
Definition	The amount of surface a shape covers	The space inside a rectangle
Measurement	Always measured in square units	cm ² , m ² , km ²
Visual	Number of unit squares that fit inside a shape	How many 1cm×1cm squares fit inside
2D Concept	Area is a 2-dimensional measure	Only applies to flat shapes

Understanding the Concept

Shape	Area Means	Visual
Square	Number of 1×1 squares inside	$a \times a$
Rectangle	Number of unit squares inside	$\text{length} \times \text{breadth}$
Triangle	Half the rectangle that contains it	$\frac{1}{2} \times \text{base} \times \text{height}$
Parallelogram	Same as rectangle with same base and height	$\text{base} \times \text{height}$
Circle	Number of unit squares inside	πr^2

Key Understanding: Area is how much space a shape covers. Think of it as "filling" the shape with unit squares.

2 Understanding the Square and Rectangle

Shape	Formula	Why It Works
Square	$A = a^2$	A square of side a can be divided into $a \times a$ small squares
Rectangle	$A = l \times b$	A rectangle of length l and breadth b can be divided into $l \times b$ small squares

Square of side 3:
[■■■] Each small square = 1 unit²
[■■■] Total squares = $3 \times 3 = 9$
[■■■] Area = 9 unit²

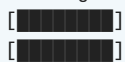
Rectangle of length 4, breadth 2:
[■■■■] Each small square = 1 unit²
[■■■■] Total squares = $4 \times 2 = 8$
Area = 8 unit²

Key Understanding: Area of rectangle = number of unit squares that fit inside.

3 Understanding the Triangle

Concept	Explanation	Example
Relation to Rectangle	A triangle is HALF of a rectangle	Triangle with same base and height = half the rectangle
Why $\frac{1}{2}$	The rectangle can be split into TWO equal triangles	Each triangle has same base and height
Height	Must be PERPENDICULAR to the base	Measured at 90° angle

Rectangle:

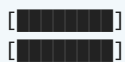
 Area = base $\times$ height = $6 \times 4 = 24$

Triangle:

 Area = $\frac{1}{2} \times$ base $\times$ height = $\frac{1}{2} \times 6 \times 4 = 12$

Why Triangle Is Half

Draw a diagonal in a rectangle:

 The diagonal splits the rectangle into
 TWO equal triangles

Each triangle has:

- Same base as rectangle
- Same height as rectangle
- Area = $\frac{1}{2} \times$ base $\times$ height

Key Understanding: A triangle is always half of a rectangle with the same base and height.

4 Understanding the Parallelogram

Concept	Explanation	Example
Definition	A quadrilateral with TWO pairs of parallel sides	Opposite sides are equal and parallel
Base	Any side can be the base	Usually the bottom side
Height	Perpendicular distance between base and opposite side	Must be measured at 90°

Parallelogram:

 Area = base $\times$ height

Rectangle:

 Same base and height = SAME area!

Why? Cut a triangle from one side and paste on the other:

 $\rightarrow$ 

Activity Understanding

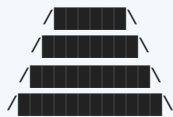
1. Draw a rectangle with length a and height h
2. Cut off a triangle from one side
3. Paste it on the other side
4. You get a parallelogram
5. Area remains the SAME!

Key Understanding: A parallelogram has the SAME area as a rectangle with the same base and height.

5 Understanding the Trapezium

Concept	Explanation	Example
Definition	A quadrilateral with ONE pair of parallel sides	Only one pair of opposite sides are parallel
Why Formula Works	Split into TWO triangles by drawing a diagonal	Area = Area of triangle 1 + Area of triangle 2
Height	Perpendicular distance between parallel sides	Must be measured at 90°

Trapezium:



$$\text{Area} = \frac{1}{2} \times (a + b) \times h$$

Split by diagonal:



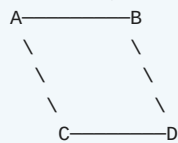
$$\text{Triangle 1: } \frac{1}{2} \times a \times h$$

$$\text{Triangle 2: } \frac{1}{2} \times b \times h$$

$$\text{Total: } \frac{1}{2} \times (a + b) \times h$$

Understanding the Split

Draw diagonal AC:



$$\text{Triangle ABC: area} = \frac{1}{2} \times AB \times h$$

$$\text{Triangle ACD: area} = \frac{1}{2} \times DC \times h$$

$$\text{Total} = \frac{1}{2} \times (AB + DC) \times h$$

Key Understanding: A trapezium is two triangles put together, sharing the same height.

6 Understanding the Circle

Concept	Explanation	Example
Relation to Rectangle	Circle can be divided into sectors, rearranged into approximate rectangle	Area = πr^2
Radius (r)	Distance from center to edge	Half the diameter
π (Pi)	Constant = $3.142 = 22/7$	Ratio of circumference to diameter

Circle divided into sectors:


 → Rearranged into approximate rectangle
 Length = πr (half circumference)
 Width = r
 Area = $\pi r \times r = \pi r^2$

Understanding π

Circumference = $\pi \times \text{diameter}$
 Area = $\pi \times \text{radius} \times \text{radius}$

Think of π as "3.14 times"
 Circle area = $3.14 \times r^2$

Key Understanding: A circle's area is about 3.14 times the square of its radius.

7 Understanding the Relationship: Shapes and Units

Shape	Area Unit	Example
Square	cm^2, m^2	A square of side 5 cm has area 25 cm^2
Rectangle	cm^2, m^2	A rectangle $10 \text{ cm} \times 6 \text{ cm}$ has area 60 cm^2
Triangle	cm^2, m^2	A triangle base 12 cm, height 5 cm has area 30 cm^2
Parallelogram	cm^2, m^2	A parallelogram base 10 cm, height 5 cm has area 50 cm^2
Trapezium	cm^2, m^2	A trapezium parallel sides 11 and 6, height 8 has area 68 cm^2
Circle	cm^2, m^2	A circle radius 7 cm has area 154 cm^2

Understanding Square Units

1 cm^2 = area of a square of side 1 cm
 1 m^2 = area of a square of side 1 m
 1 km^2 = area of a square of side 1 km

$1 \text{ m}^2 = 10,000 \text{ cm}^2$
 $1 \text{ km}^2 = 1,000,000 \text{ m}^2$

8 Common Misunderstandings to Avoid

Misunderstanding	Why It's Wrong	Correct Understanding
"Triangle = base $\times$ height"	Missing the $\frac{1}{2}$	Triangle = $\frac{1}{2} \times \text{base} \times \text{height}$
"Height = side length"	Height must be perpendicular	Height is perpendicular distance
"All sides need to be known"	Only base and height needed for triangle/parallelogram	Perimeter needs all sides, area needs base and height
" $\pi = 3$ "	π is 3.142 or 22/7	Use the exact value given in the problem
"Trapezium has two pairs of parallel sides"	Only ONE pair	Parallelogram has TWO pairs

9 Understanding the Relationship: Perimeter vs Area

Concept	Perimeter	Area
What it measures	Distance around the shape	Space inside the shape
Units	cm, m, km (linear)	cm ² , m ² , km ² (square)
Triangle	Sum of all 3 sides	$\frac{1}{2} \times \text{base} \times \text{height}$
Rectangle	$2 \times (l + b)$	$l \times b$
Square	$4 \times a$	a^2
Circle	$2\pi r$ (circumference)	πr^2

Understanding the Difference

Square of side 4:

Perimeter = $4 + 4 + 4 + 4 = 16$ cm

Area = $4 \times 4 = 16$ cm²

Rectangle of length 6, breadth 3:

Perimeter = $2 \times (6 + 3) = 18$ cm

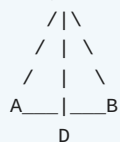
Area = $6 \times 3 = 18$ cm²

Different shapes can have same area but different perimeters!

10 Understanding the Relationship: Base and Height

Shape	Base	Height
Triangle	Any side	Perpendicular distance from opposite vertex to that side
Parallelogram	Any side	Perpendicular distance between that side and its parallel side
Trapezium	The parallel sides	Perpendicular distance between the parallel sides

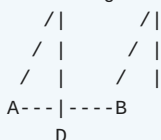
Triangle:



Base = BC

Height = AD (perpendicular from A to BC)

Parallelogram:



Base = AB

Height = DE (perpendicular from D to AB)

Key Understanding: The height is ALWAYS perpendicular to the base, not the slanted side!

11 Check Your Understanding

No.	Question	Explain
1	What is area?	The amount of space inside a shape
2	Why is triangle area half of rectangle?	It's half of a rectangle with same base and height
3	Why is height perpendicular to base?	Area measures "straight up" space, not slanted
4	Why is parallelogram area same as rectangle?	It can be rearranged to form a rectangle
5	Why is trapezium formula $\frac{1}{2} \times (\text{sum of parallel sides}) \times \text{height}$?	It's two triangles with same height
6	What does π represent in circle area?	The constant 3.142 or $22/7$
7	Why is area in square units?	It measures number of unit squares inside
8	What is the difference between area and perimeter?	Area = inside space, Perimeter = distance around
9	Why can any side be the base of a parallelogram?	Any side has a corresponding height
10	Why is the circle area formula πr^2 ?	Circle rearranged to approximate rectangle of length πr and width r

12 Key Understandings to Remember

- AREA MEASURES THE SPACE INSIDE A SHAPE.
- AREA IS ALWAYS IN SQUARE UNITS.
- THE HEIGHT IS ALWAYS PERPENDICULAR TO THE BASE.
- A TRIANGLE IS HALF A RECTANGLE.
Area = $\frac{1}{2} \times \text{base} \times \text{height}$.
- A PARALLELOGRAM = A RECTANGLE WITH SAME BASE AND HEIGHT.
Area = $\text{base} \times \text{height}$.
- A TRAPEZIUM = TWO TRIANGLES WITH SAME HEIGHT.
Area = $\frac{1}{2} \times (\text{sum of parallel sides}) \times \text{height}$.
- A CIRCLE = APPROXIMATELY A RECTANGLE.
Area = πr^2 .
- $\pi = 3.142 = 22/7$.

Revision Tip: Visualize each shape and how it relates to others. Understanding "why" formulas work helps you remember them better!